

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

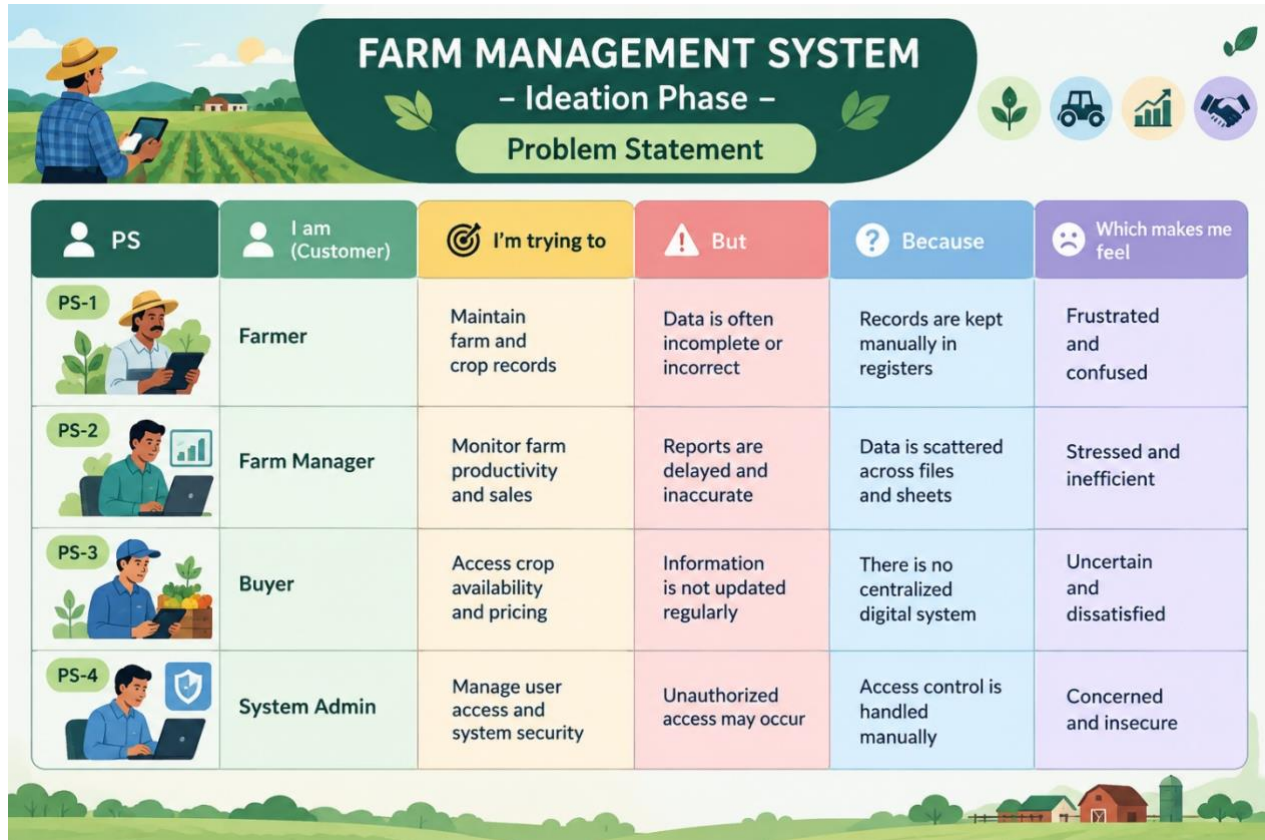
The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

1. IDEATION PHASE

1.2 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure



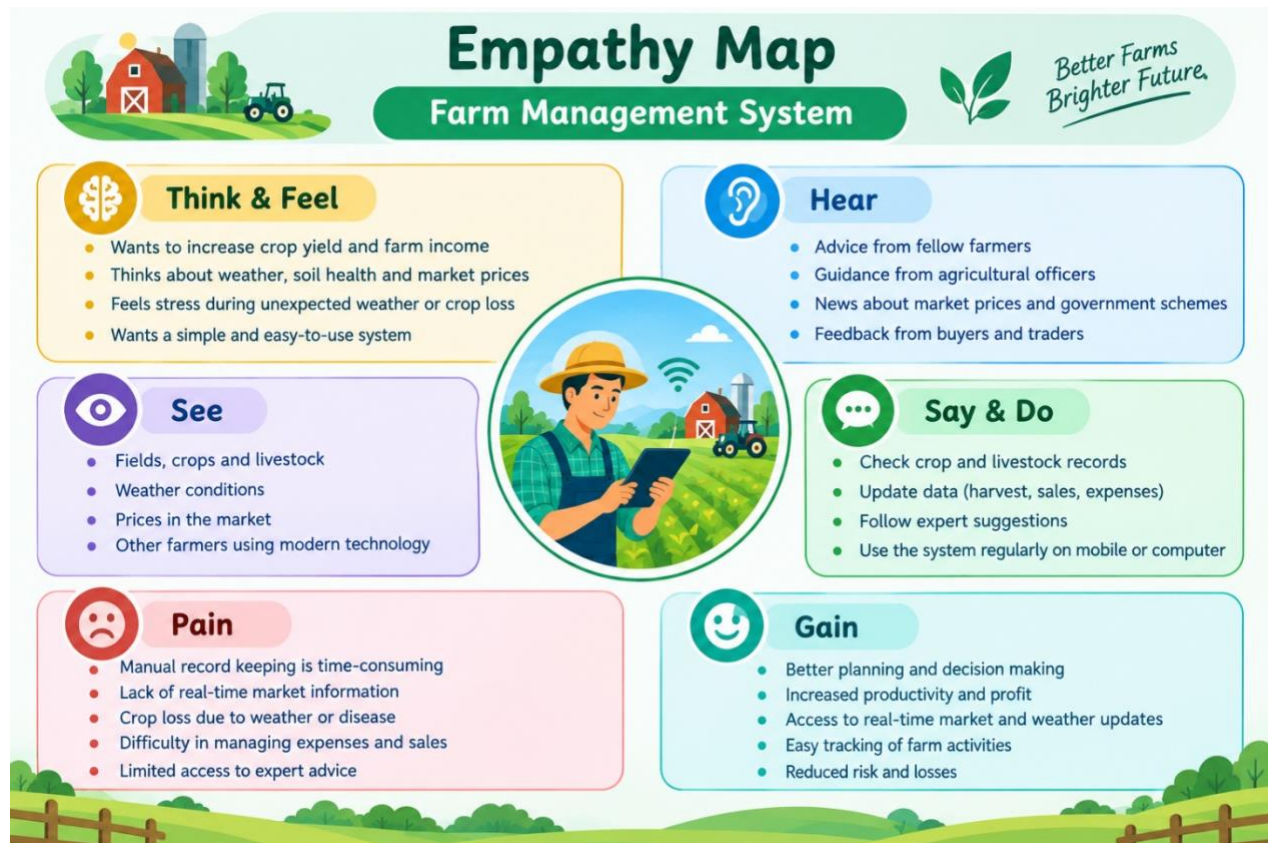
1.3 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



1.4 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.



2. REQUIREMENT ANALYSIS PHASE

1.5 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



1.6 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6	Dashboard Visualization
FR-7	User Role Management

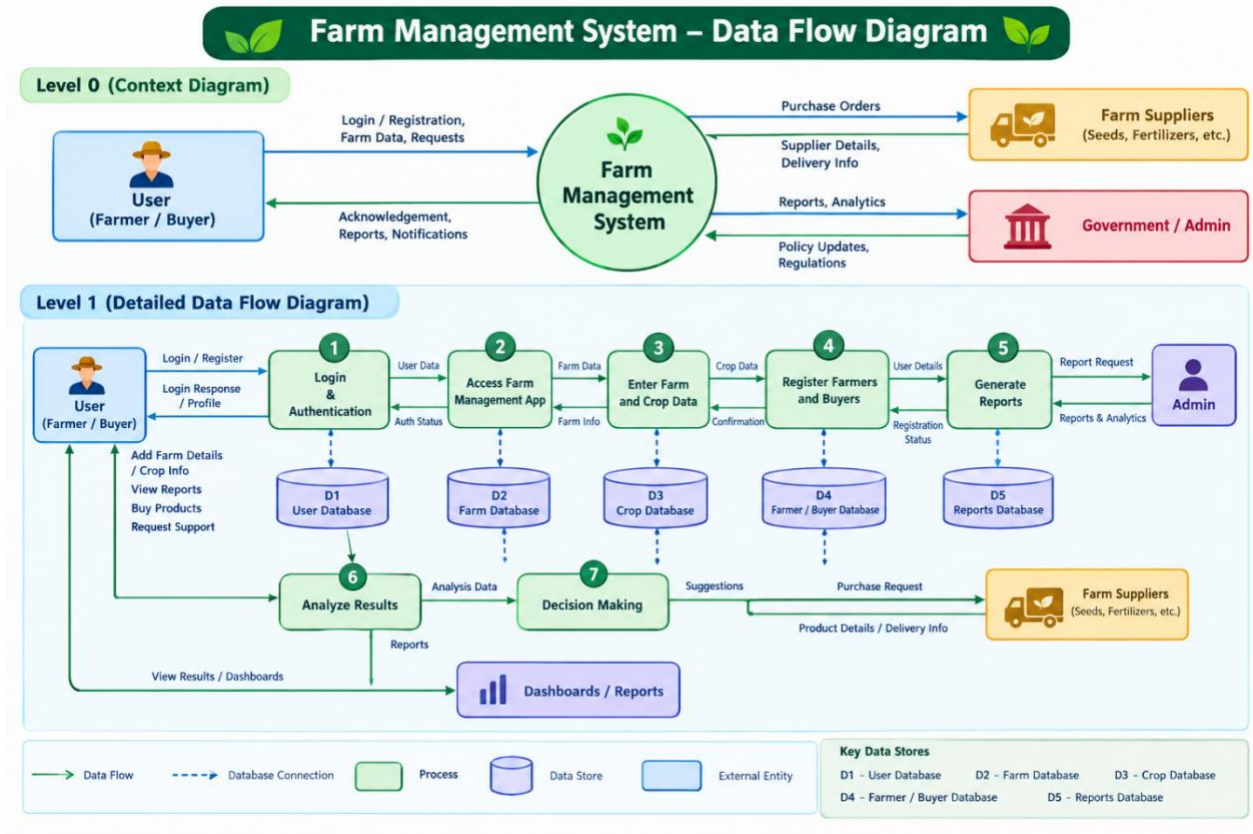
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

1.7 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



1.8 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

3.PROJECT DESIGN PHASE

1.9 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



1.10 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

1.11 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



1. Project Planning

-  **Create Salesforce Dev Org**
Set up development environment and configure basic settings.
 -  **Develop Custom Objects**
Create objects for Farm, Crop, Farmer, Buyer, etc.
 -  **Configure Tabs & Lightning App**
Add tabs, custom app and navigation.
 -  **Create Fields & Relationships**
Define fields, set relationships between objects.
 -  **Design Profiles & Roles**
Assign permissions and access controls.
 -  **Create Users & Sharing Rules**
Manage user access and record sharing.
 -  **Develop Reports & Dashboards**
Build analytical views for better insights.
 -  **Conduct Testing & Reviews**
Verify functionality and make improvements.



3. Estimation

Work Item	Estimated Effort (Days)
Setup Dev Org & Configurations	2
Custom Objects & Fields	5
Relationships & Tabs	3
Profiles, Roles & Sharing Rules	2
Reports & Dashboards	3
Testing & Deployment	2

 Total Estimated Effort : 17 Days

2. Project Schedule

Planning & Setup	Week 1
Development	Week 2 – 3
Testing	Week 4
Review & Bug Fixes	Week 5
Deployment	Week 6

 Sprint Schedule

Sprint	Duration	Focus Area
Sprint 1	1 Week	Setup & Basic Configuration
Sprint 2	1 Week	Custom Objects & Fields
Sprint 3	1 Week	UI & Lightning App
Sprint 4	1 Week	Reports & Dashboards
Sprint 5	1 Week	Testing & Deployment

 Salesforce Platform | Farm Management System
Better Farming | Smarter Decisions

4.PROJECT PLANNING & SCHEDULING

1.12 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Use the below template to create product backlog and sprint schedule

sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

5 .PROJECT DEVELOPMENT PHASE:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

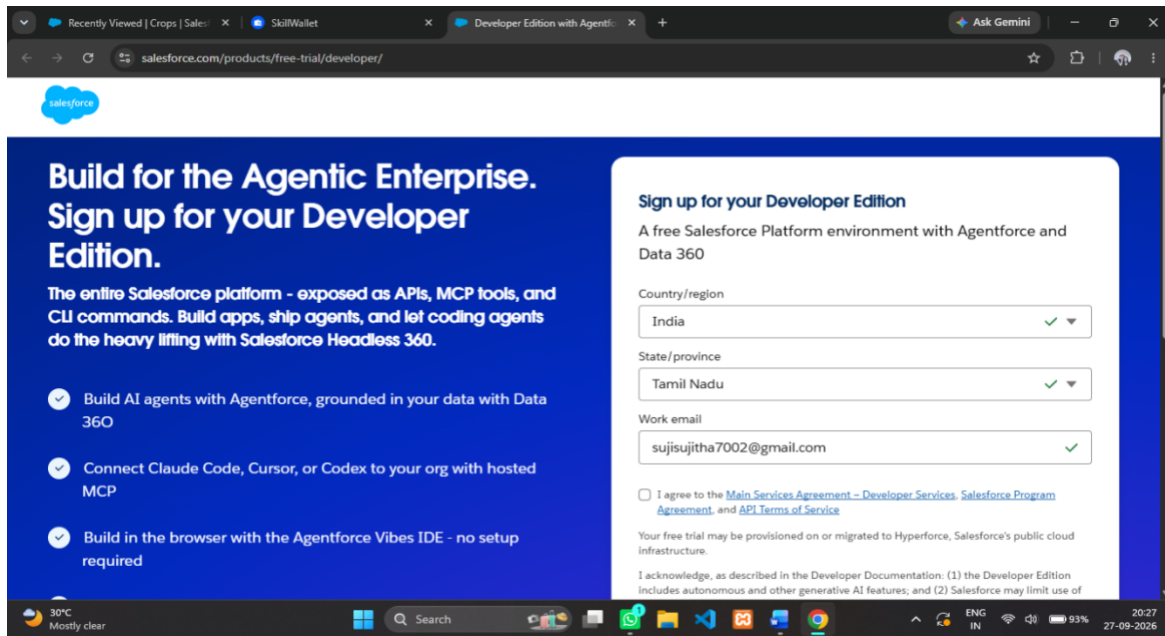
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will receive a mail from Salesforce and by using the verification account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



The screenshot shows a web browser window with the URL salesforce.com/products/free-trial/developer/. The page has a blue header with the Salesforce logo. The main content area is split into two columns. The left column has a dark blue background with white text that reads: "Build for the Agentic Enterprise. Sign up for your Developer Edition. The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360." Below this are three bullet points with checkmarks: "Build AI agents with Agentforce, grounded in your data with Data 360", "Connect Claude Code, Cursor, or Codex to your org with hosted MCP", and "Build in the browser with the Agentforce Vibes IDE - no setup required". The right column has a white background with a blue border. It contains the heading "Sign up for your Developer Edition" and the subtext "A free Salesforce Platform environment with Agentforce and Data 360". Below this are three form fields: "Country/region" with a dropdown menu showing "India", "State/province" with a dropdown menu showing "Tamil Nadu", and "Work email" with a text input showing "sujisujitha7002@gmail.com". Each field has a green checkmark icon on the right. Below the form fields is a checkbox labeled "I agree to the Main Services Agreement - Developer Services, Salesforce Program Agreement, and API Terms of Service". At the bottom of the right column, there is a disclaimer: "Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of". The browser's address bar shows the URL and the page title "Developer Edition with Agentforce". The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray at the bottom right shows the date and time "20:27 27-09-2026" and the battery level "93%".

Build for the Agentic Enterprise.
Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required

Sign up for your Developer Edition
A free Salesforce Platform environment with Agentforce and Data 360

Country/region
India ✓

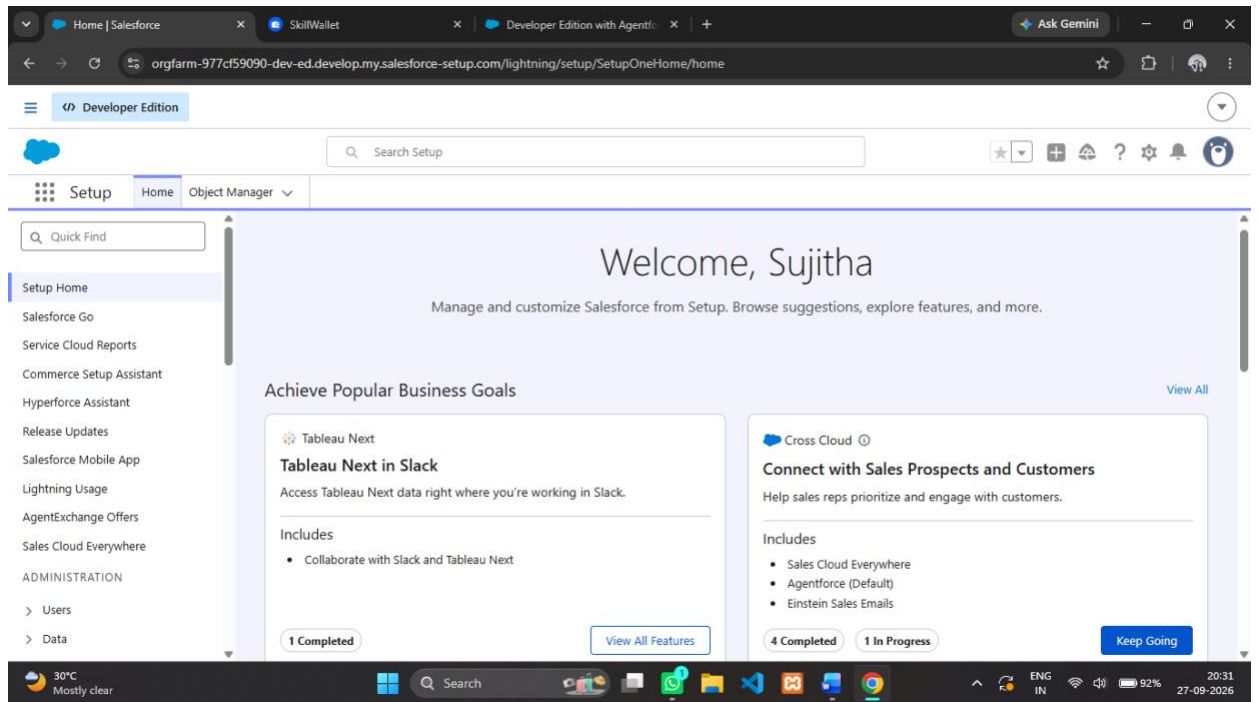
State/province
Tamil Nadu ✓

Work email
sujisujitha7002@gmail.com ✓

☐ I agree to the [Main Services Agreement - Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.
I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

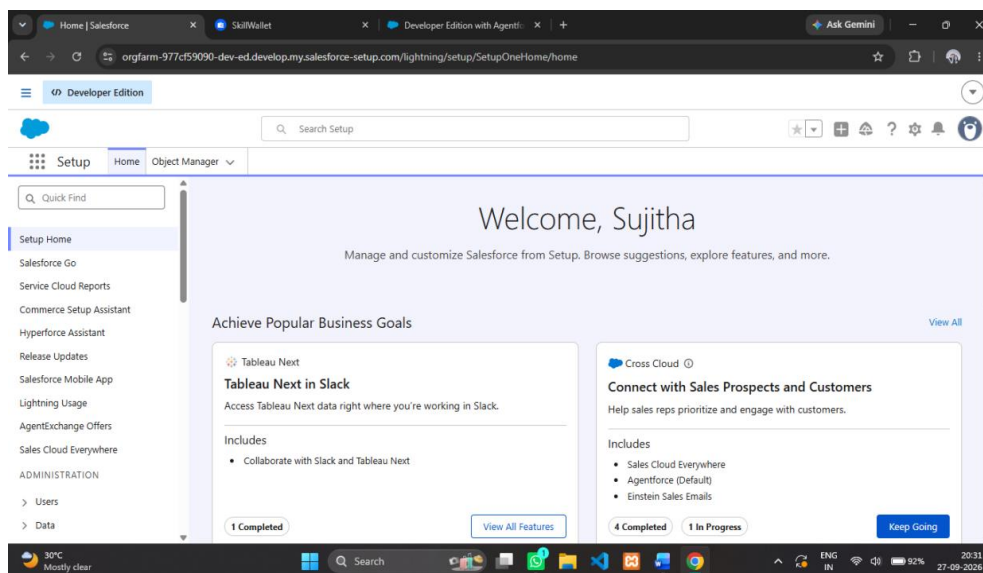
1. Farm
2. Crop
3. Farmer
4. Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

The screenshot displays the Salesforce Setup interface in the Developer Edition. The browser address bar shows the URL: `orgfarm.977c59090-dev-ed.develop.my.salesforce.com/lightning/setup/ObjectManager/011a00000FMGg/edit?address=%2F011a00000FMGg%2F%3FuetL...`. The navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar shows the 'Farm' object selected under 'SETUP > OBJECT MANAGER'. The main content area is titled 'Custom Object Definition Edit' and contains the following fields:

- Label:** Farm (with a red asterisk indicating it is required)
- Plural Label:** Farms (with a red asterisk indicating it is required)
- Starts with vowel sound:** ☐
- Object Name:** Farm (with a red asterisk indicating it is required)
- Description:** (empty text area)

At the bottom of the page, there is a link to 'Open the standard Salesforce.com Help & Training window'.

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

The screenshot shows the Salesforce Setup page in the 'Developer Edition' environment. The 'Object Manager' tab is selected, and the 'Crop' object is being defined. The 'Details' section on the left lists various configuration options. The main area shows the 'Label' as 'Crop', 'Plural Label' as 'Crops', and 'Object Name' as 'Crop'. The 'Description' field is empty. The 'Context-Sensitive Help Setting' is set to 'Open the standard Salesforce.com Help & Training window'. The 'Content Name' is set to 'None'. The 'Enter Record Name Label and Format' section is at the bottom.

Setup > OBJECT MANAGER
Crop

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookups Filters

Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label: Example: Account

Plural Label: Example: Accounts

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name: Example: Account

Description:

Context-Sensitive Help Setting: ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name:

Enter Record Name Label and Format

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

The screenshot displays the Salesforce Setup interface. The top navigation bar includes the 'Developer Edition' tab and a search bar. The left sidebar shows the 'Setup' menu with 'Object Manager' selected. The main content area is titled 'Farmer' and shows the 'Custom Object Definition Edit' page. The 'Custom Object Information' section contains the following fields:

- Label:** Farmer (with a red error message 'Required Information')
- Plural Label:** Farmers
- Starts with vowel sound:** ☐
- Object Name:** Farmer
- Description:** (empty text area)

The page also includes a 'Help for this Page' link and a 'Save' button. The bottom status bar shows the system time as 20:38 on 27-09-2026.

- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3

Tabs

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

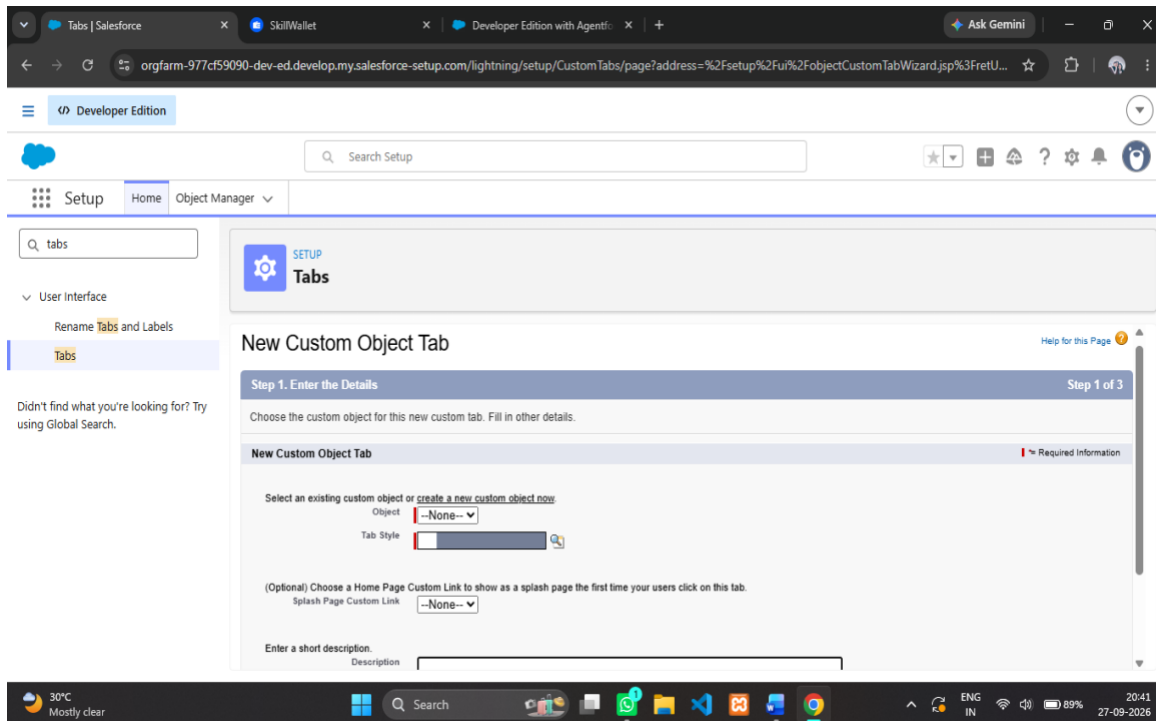
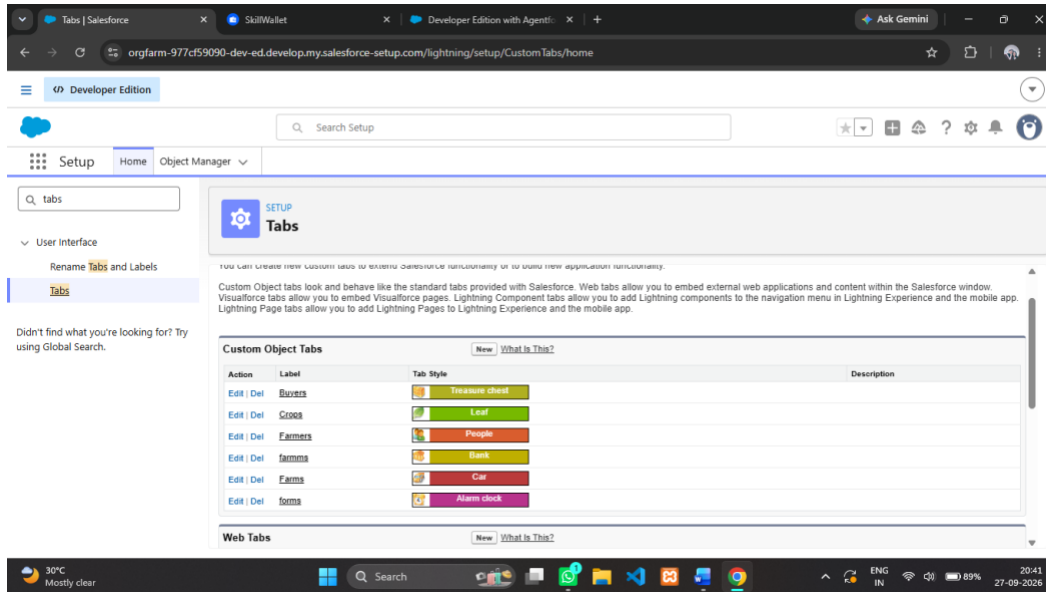
Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farm.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.



Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

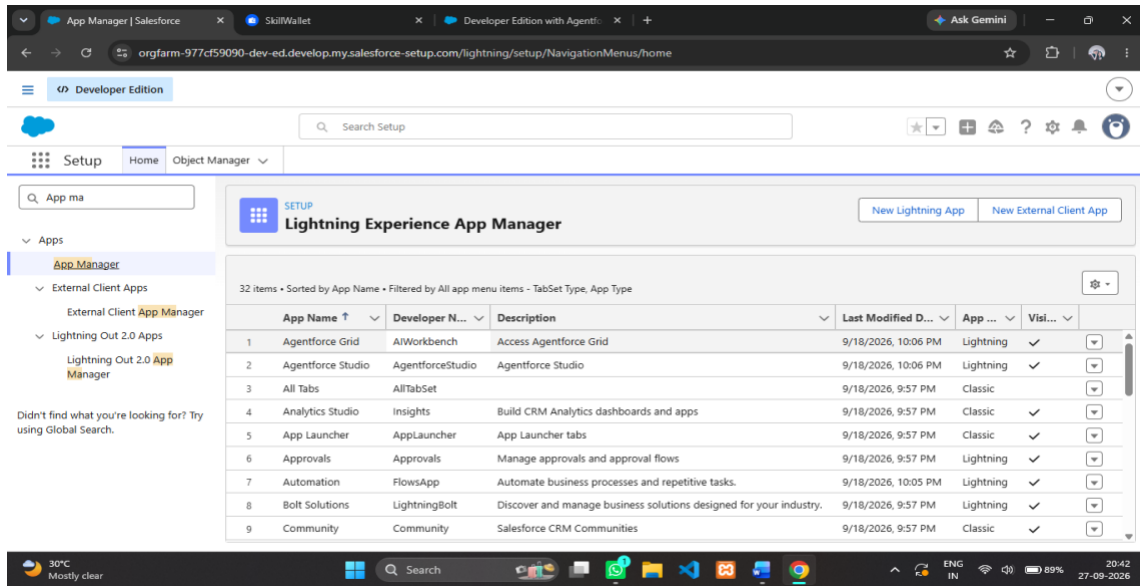
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

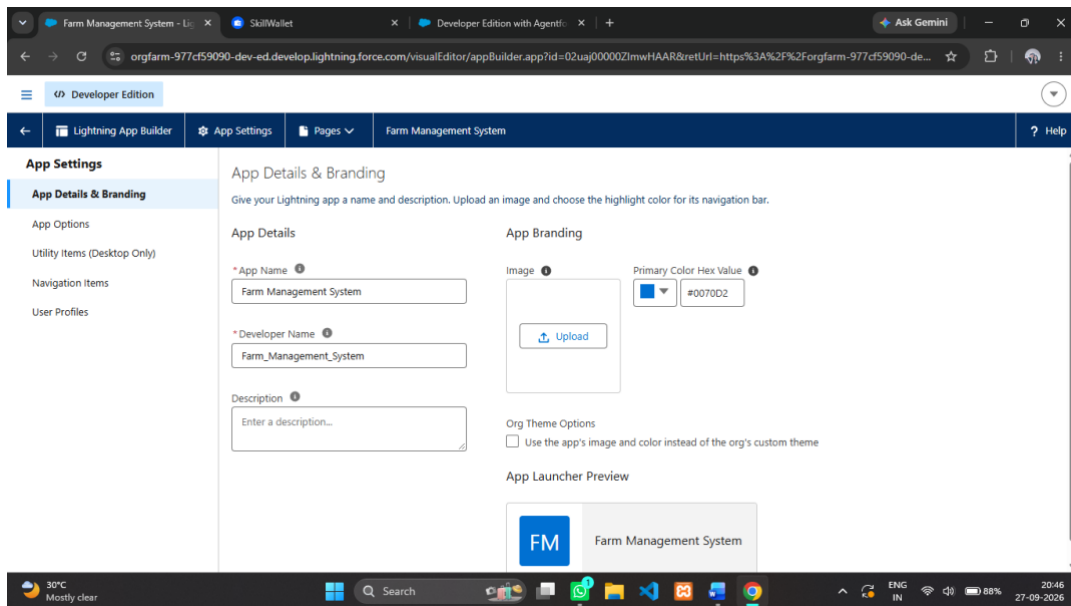
Create the Farm Management System app



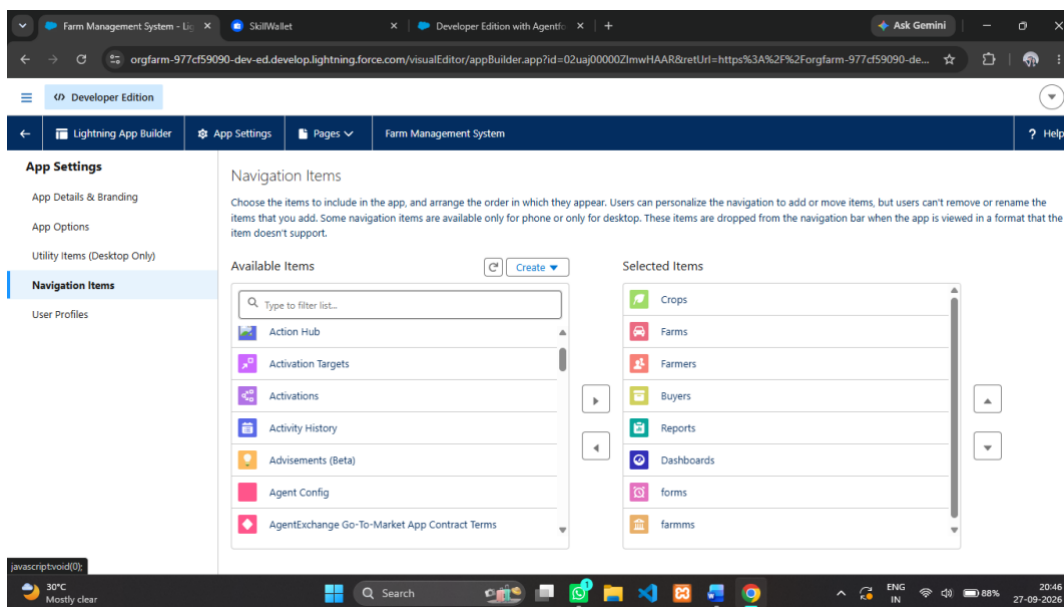
- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
- From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



4. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

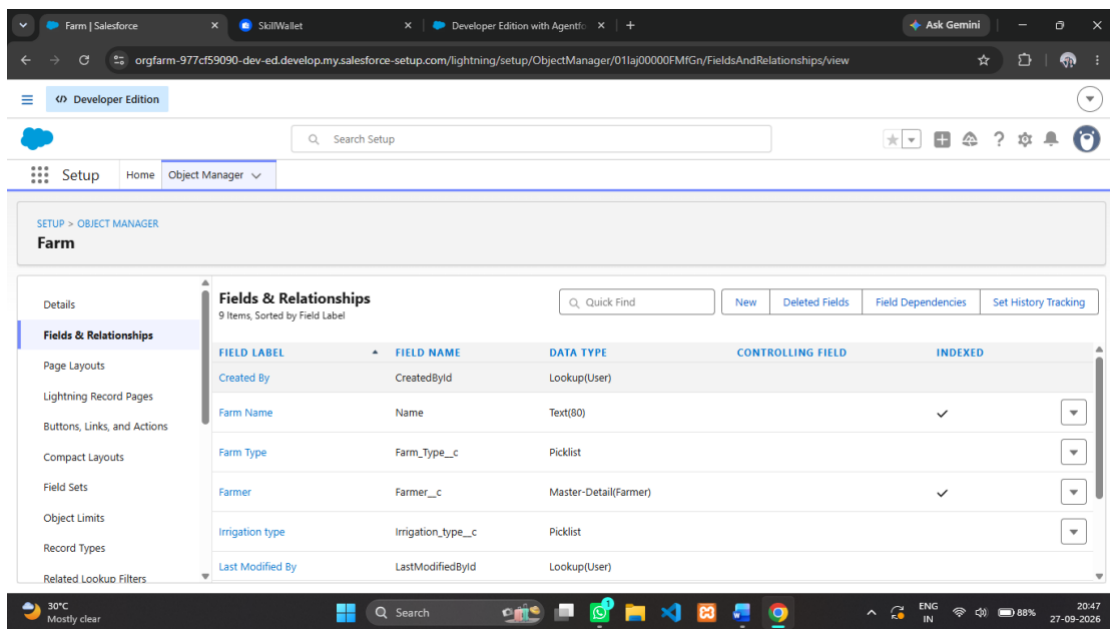
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

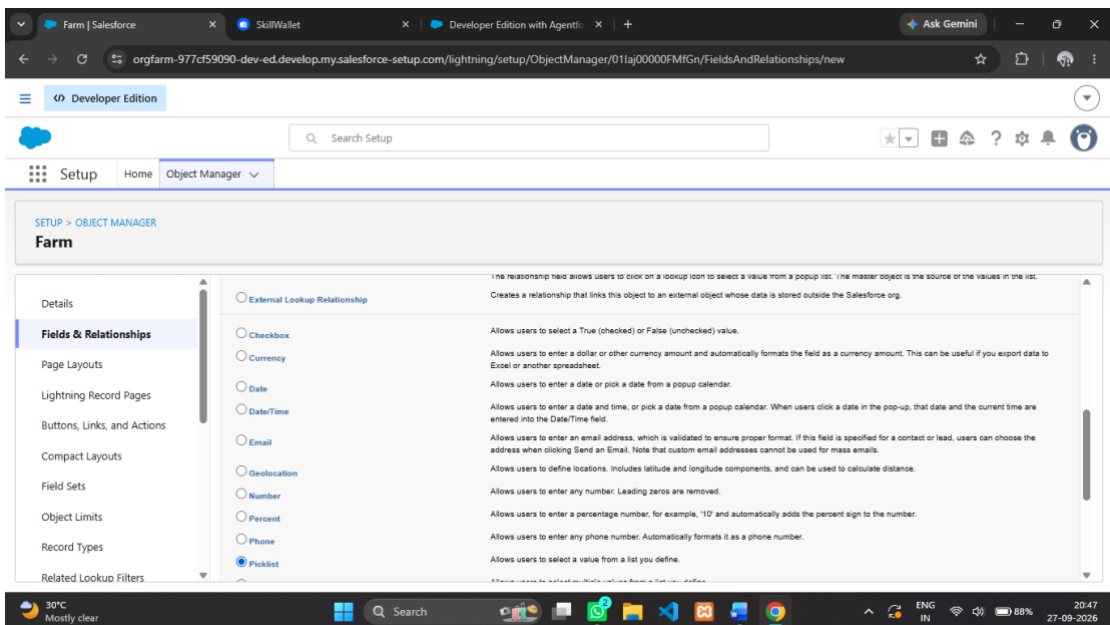
Activity 1:

Creation of fields for the Farm objects:

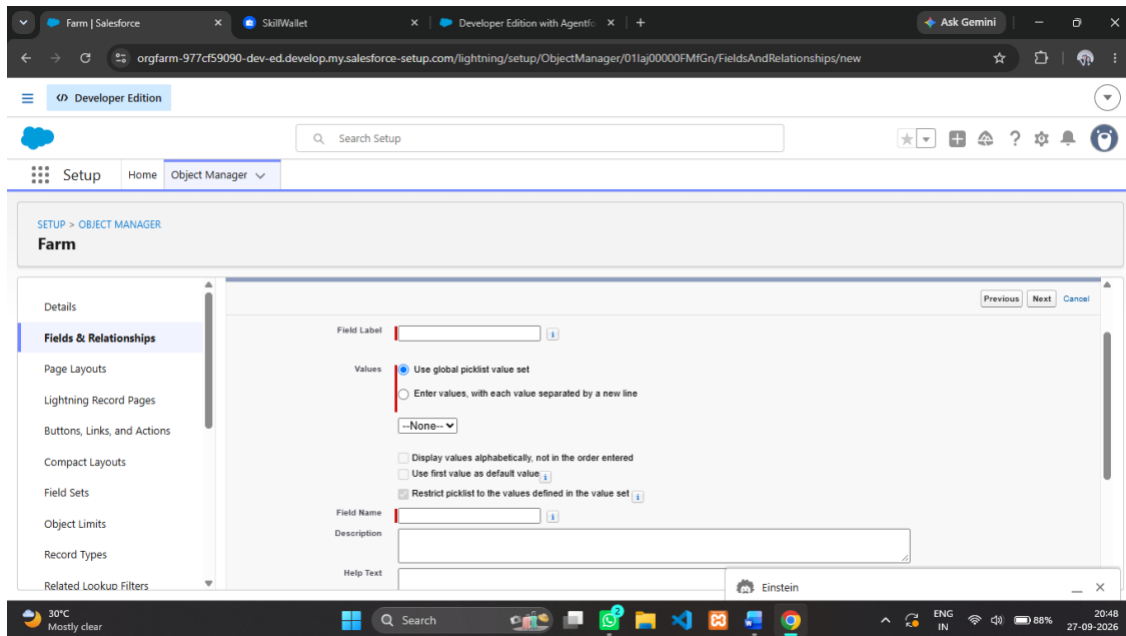
1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



1. Choose Data type as a picklist, Click next



1. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
2. Next, Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

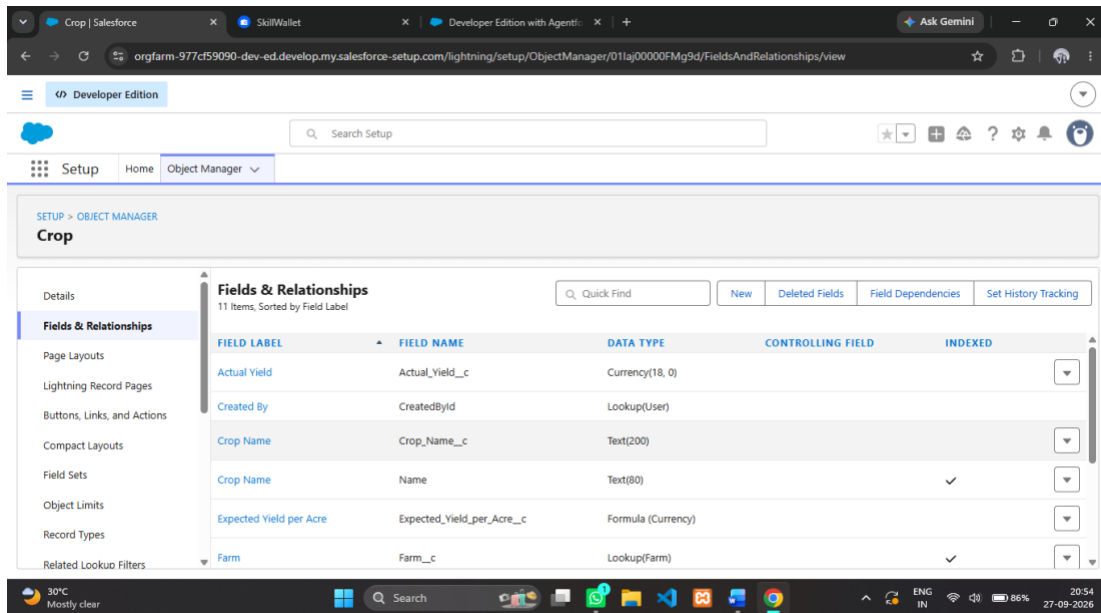
1. Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as a text, enter field label (Location)
4. Choose data type as text, enter field label (size)
5. Create a Master Detail relationship with Farmer Object.

Activity 2:

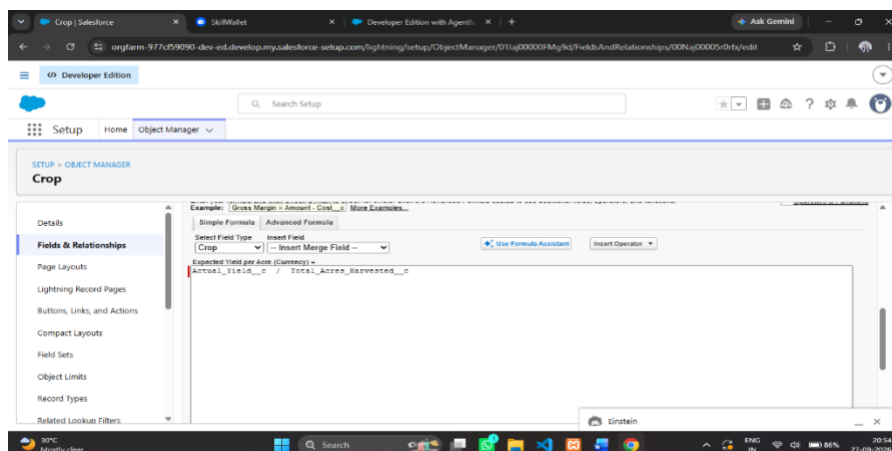
Creation of fields for the Crop objects:

1. Choose Data type As a text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date, Enter Field label (Planting Date), then save.
3. Again we select data type as a date, Enter field label (Harvest date), save
4. Next We Create Lookup relationship on crop object related to Farm Object.

5. We select a data type as currency, Enter field Label (Actual Yield), Next Save it.
6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length(20) and save.
7. Now We create Formula Field For That Select data Type As a Formula, Click Next.



8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.
9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

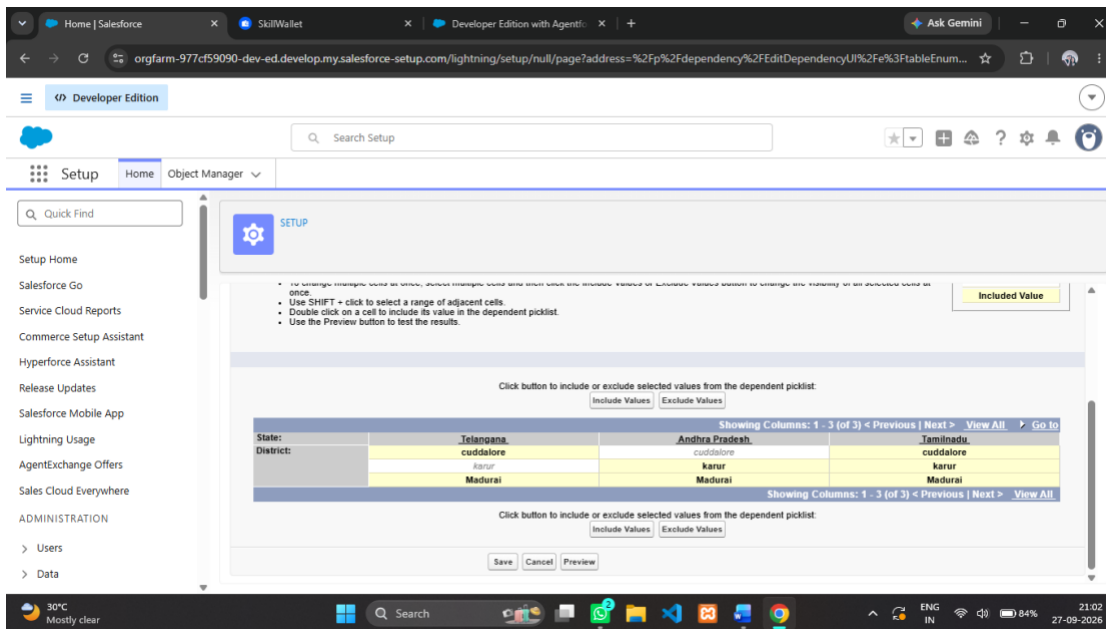
1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

The screenshot shows the Salesforce Developer Edition interface. The browser address bar displays the URL: `orgfarm-977cd59090-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01tj00000FMgHh/FieldsAndRelationships/new`. The page title is "Farmer". The left sidebar shows the "Object Manager" menu with "Farmer" selected. The main content area displays the "New Custom Field" wizard for the "Farmer" object. The wizard is at "Step 2: Enter the details". The "Field Label" is "Contact Number". The "Length" is "10" and "Decimal Places" is "0". The "Field Name" is "ContactNumber". The "Description" is "Contact Number of the Farmer". The "Help Text" is "Contact Number of the Farmer". The "Previous", "Next", and "Cancel" buttons are visible at the top right of the wizard.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its cities.
6. Click Preview, then test the dependency by selecting a different State and viewing the associated Values available for Particular state.

7. Click Close to close the preview window, Click Save.



Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions,

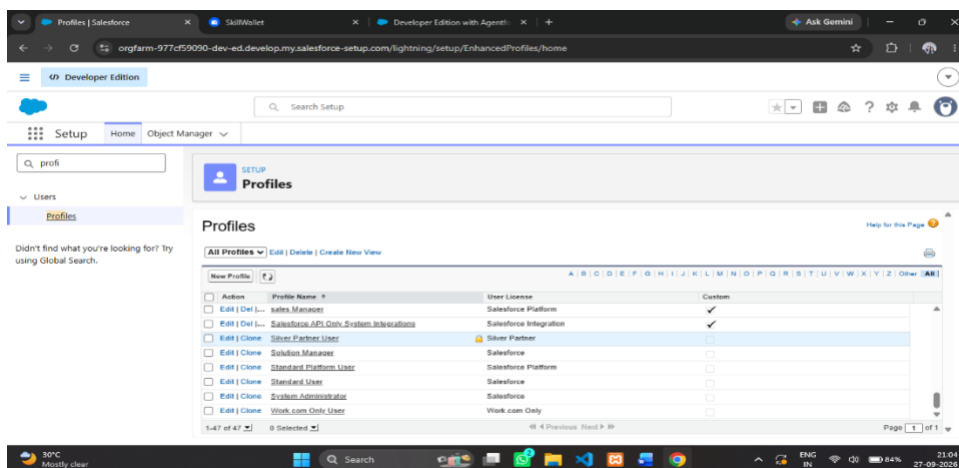
Tab ,settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.
5. 5. Click save.



Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

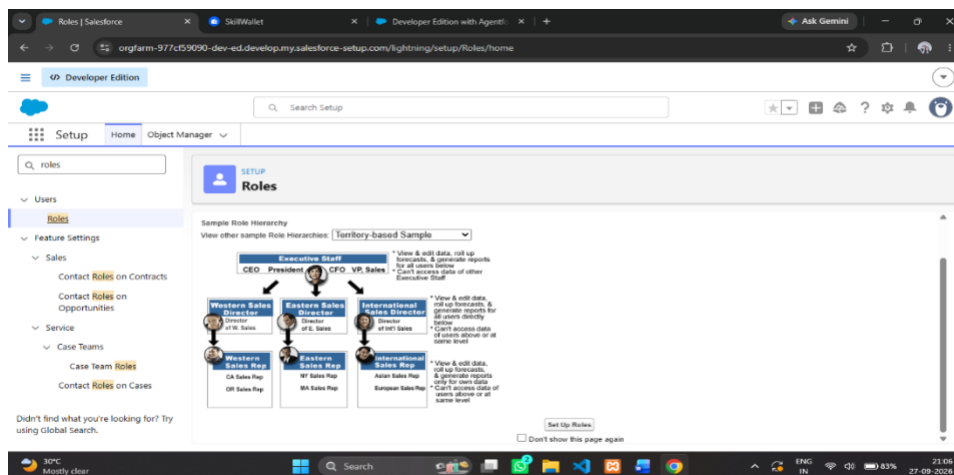
Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

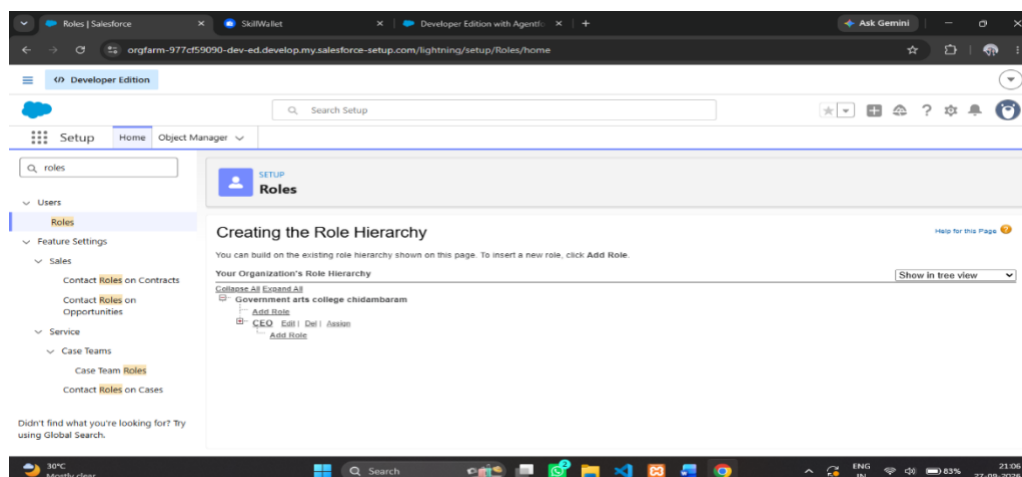
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user

account settings determine what features and records the user can access.

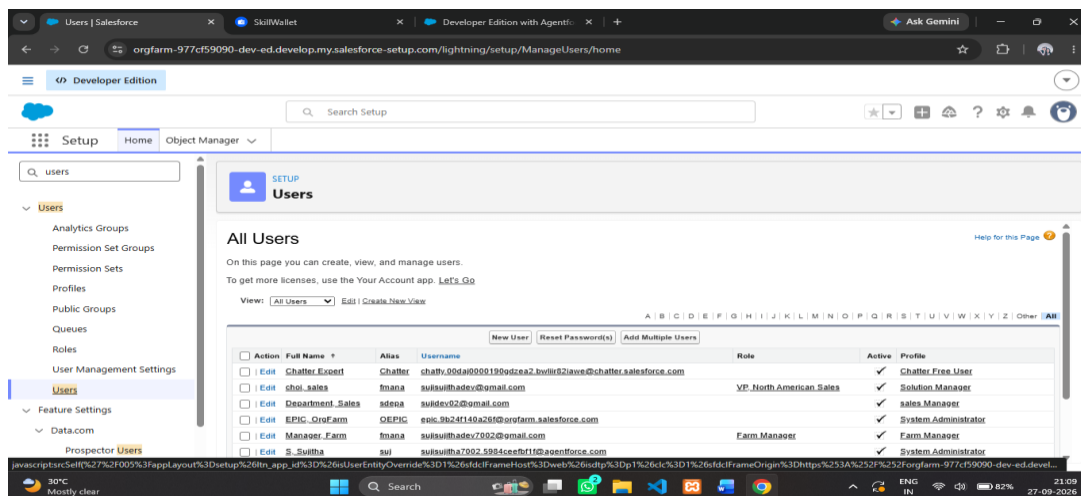
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find

salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

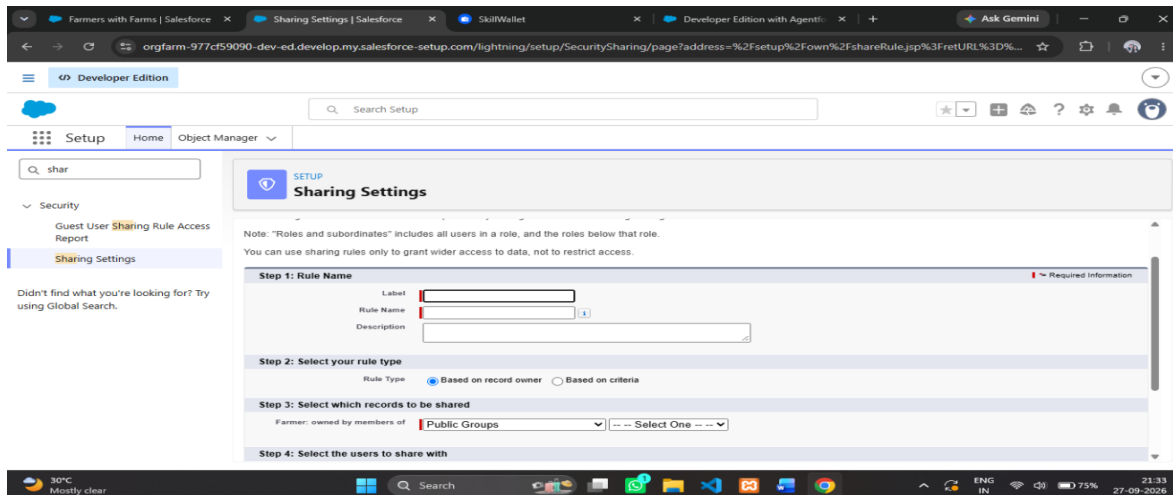
Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.



Activity 2:

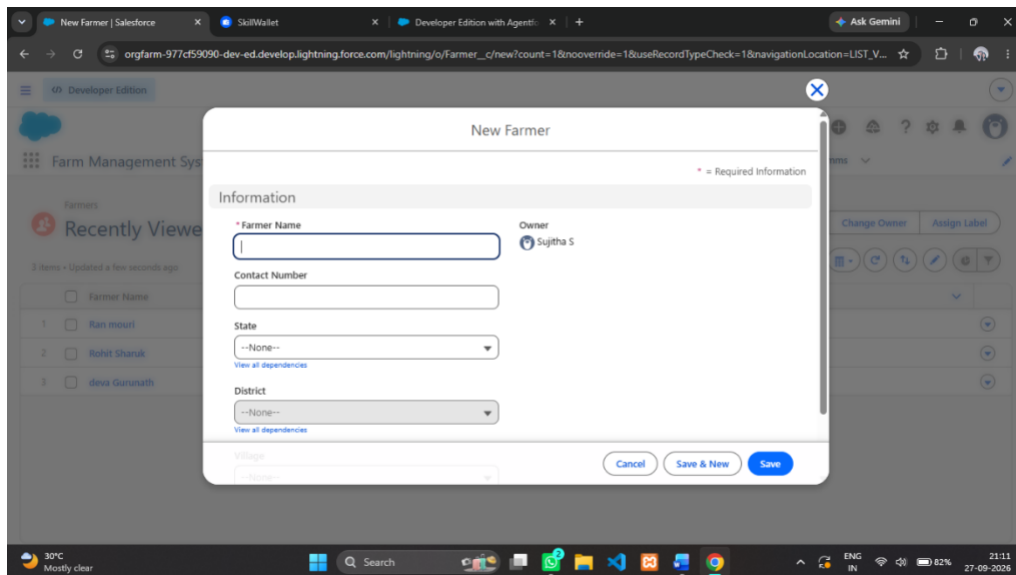
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



The screenshot shows a web browser window with the URL `orgfarm-977cf59090-dev-ed.develop.lightning.force.com/lightning/o/farmer_/c/new?count=1&nooverride=1&useRecordTypeCheck=1&navigationLocation=LIST_V...`. The page displays the 'New Farmer' form, which is a modal window with a title bar and a close button. The form is titled 'New Farmer' and includes a note '* = Required Information'. The form is divided into sections: 'Information' and 'Owner'. The 'Information' section contains fields for 'Farmer Name' (required), 'Contact Number', 'State' (dropdown menu), 'District' (dropdown menu), and 'Village' (dropdown menu). The 'Owner' section shows 'Sujitha S' as the owner. At the bottom of the form, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The background of the browser shows the Salesforce interface with the 'Farm Management System' app selected and a list of 'Recently Viewed' farmers.

Farmer Name

Contact Number

State

District

Village

Owner: Sujitha S

Buttons: Cancel, Save & New, Save

Milestone 11-Reports:

What are Reports?

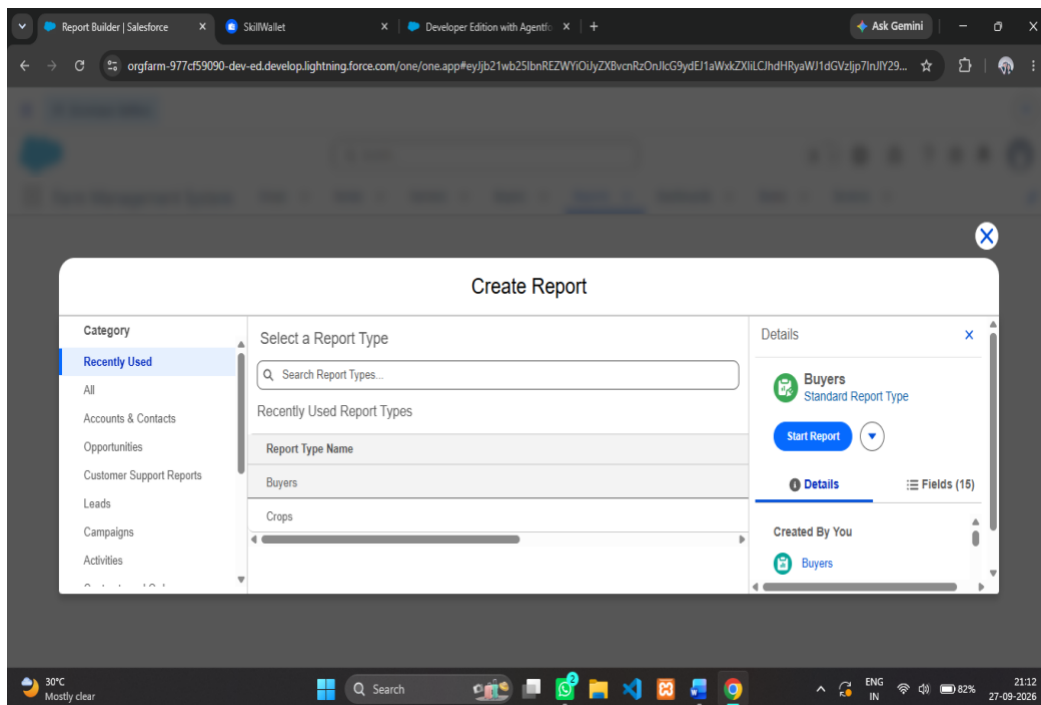
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot shows the Salesforce Reports interface. The top navigation bar includes the Salesforce logo, a search bar, and various utility icons. The main navigation menu on the left includes 'Reports', 'Dashboards', 'Forms', and 'Farms'. The 'Reports' section is active, displaying a list of recent reports. The table below shows the details of these reports.

Report Name	Description	Folder	Created By	Created On	Subscribed
Buyers		Public Reports	Sujitha S	9/26/2026, 9:47 AM	
Farmers with Farms		Public Reports	Sujitha S	9/26/2026, 9:02 AM	
Crops		Public Reports	Sujitha S	9/26/2026, 9:45 AM	

The interface also includes a sidebar on the left with options like 'Recent', 'Created by Me', 'Private Reports', 'Public Reports', 'All Reports', 'FOLDERS', 'All Folders', 'Created by Me', and 'Shared with Me'. The bottom status bar shows the system clock, temperature, and battery level.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

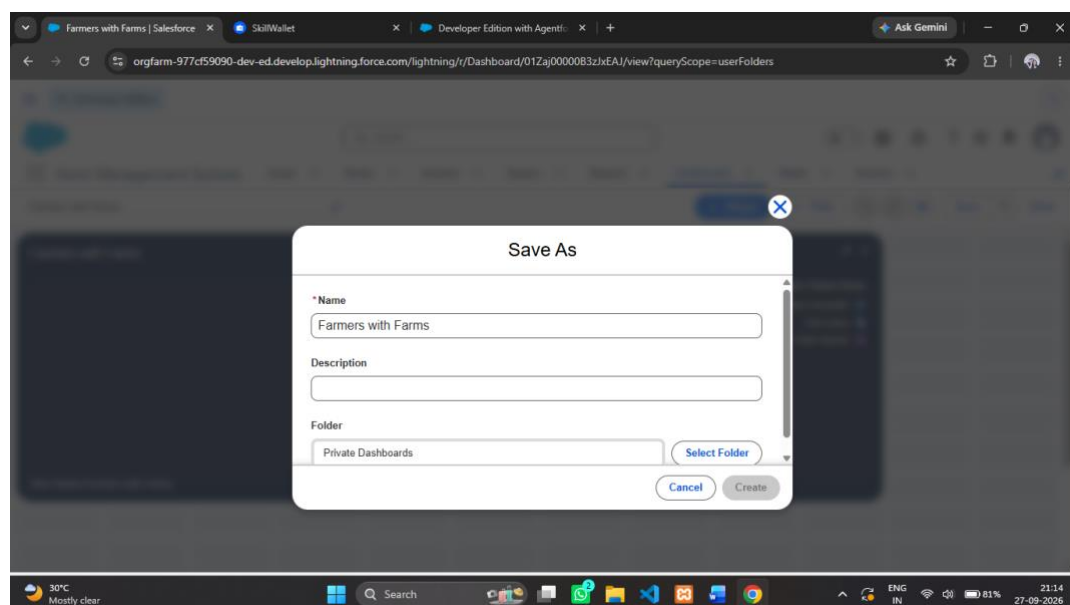
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

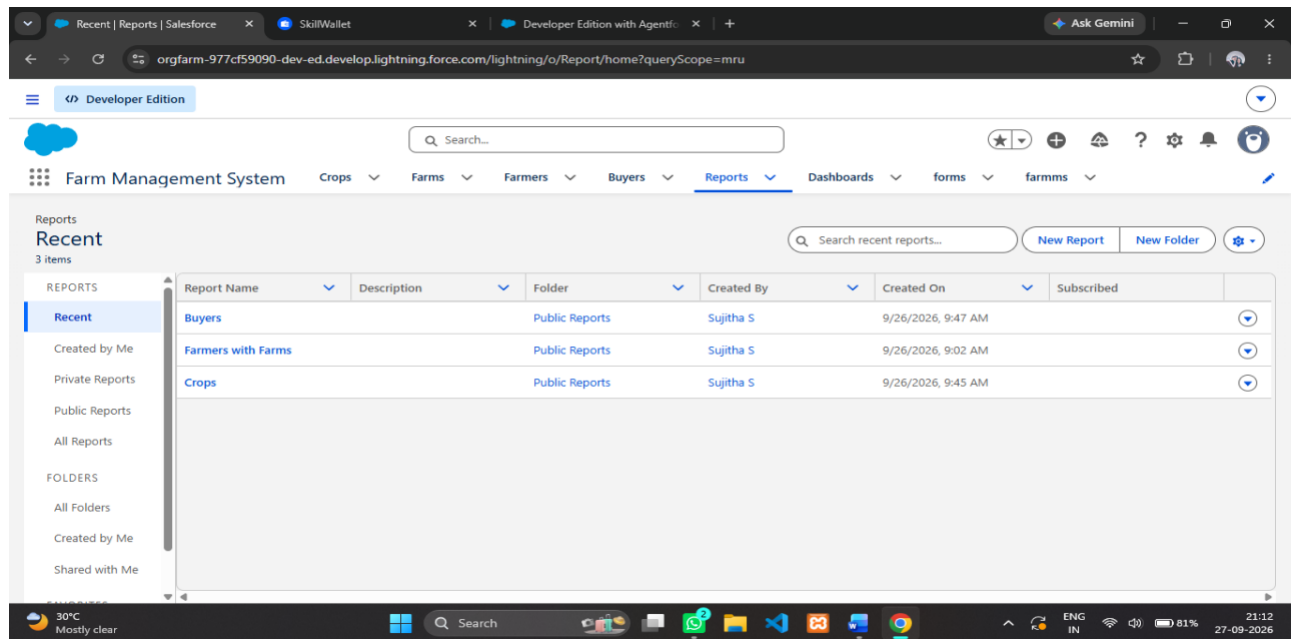
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

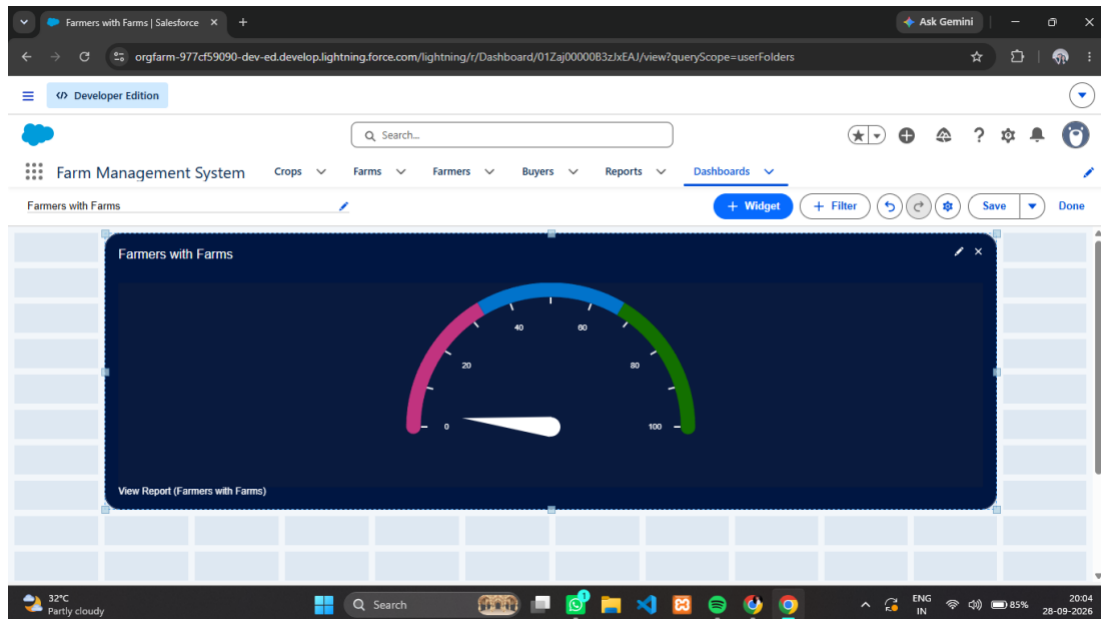
Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



6. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

7. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork

- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

8. PROJECT DEMONSTRATION

- Demonstrated the completed Farm Management System in Salesforce.
- Demonstrated the Farm, Crop, Farmer and Buyer objects.
- Demonstrated the created records and entered sample data.
- Demonstrated the fields and relationships between the objects.
- Demonstrated Profiles, Roles and User access.
- Demonstrated Sharing Rules.
- Demonstrated the Reports generated from the system.
- Demonstrated the Dashboard and its data visualization.
- Verified the overall working of the application.
- Demo Video < <https://drive.google.com/file/d/1WjFx6ttLo8iTv1lf3Y9sdrMsLc3sxa00/view?usp=drivesdk>

9. CONCLUSION

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You

